PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: Healthcare

Version: 1.0

1. Project Overview

- **Problem Statement:** Medical consultations are often inefficient because patients provide unstructured, colloquial descriptions that take doctors significant time to decipher.
- **Target Domain:** Primary healthcare clinics and Public
- **Proposed Solution Summary:** An AI-driven engine that parses unstructured human language into structured medical reports, categorizing risks and suggesting workflows for physician approval.

2. Background & Business Objective

- **Background of the Problem:** Based on Ahmad et al. (2017) "An assessment of patient waiting and consultation time in a primary healthcare clinic" (PMC5420318), in a primary care clinic serving 60 patients daily, and stated that patients waited an average of 41 minutes from registration to seeing a doctor (with only 53% registering within the 15-minute target), while the actual doctor consultation averaged 18.21 minutes (only 41.8% falling within the ideal 10–20 minute range). The study identified key bottlenecks: understaffed registration counters handling both registration and appointments, patients arriving early and clustering at the same time, and no structured system to collect patient history before the consultation. Critically, pre-consultation assessment by nurses took nearly 14 minutes time that could be repurposed for AI-driven data collection. (Ahmad, 2017)

- **Importance of Solving This Issue:** Solving the prolonged waiting time in primary healthcare clinics is critically important because long waits directly cause patient harm both physically and emotionally. As documented in the Bernama article (2024), a mother waited four hours with her toddler suffering from food poisoning, leaving her "worried about my child's health," while a 71-year-old diabetic patient reported that long waits worsened his condition, forcing him to "sleep in a wheelchair." Beyond physical deterioration, prolonged waiting increases patient anxiety, reduces trust in the public health system, and may discourage patients from seeking timely care—leading to delayed diagnoses and worse health outcomes. Therefore, reducing waiting time is not merely an efficiency metric but a patient safety and welfare imperative. (Bernama, 2025)

- **Strategic Fit / Impact:** Aligns with healthcare digital transformation and AI-assisted clinical decision support. The system can be integrated Ilmu GLM 5.1 AI to reduce waiting time and save clinician hours daily per clinic, with future collaboration potential with MOH and primary care networks.

3. Product Purpose

3.1. Main Goal of the System: To provide an AI-enabled pre-consultation and triage partner that converts unstructured patient language into structured clinical insights using the Ilmu GLM 5.1 model, reduces patient waiting time, saves clinician hours, and incorporates a doctor-in-the-loop approval mechanism before generating the final PDF summary for consultation.

3.2. Intended Users (Target Audience):

- **Patients** – input symptoms in natural language (text or voice-to-text) before seeing a doctor.
- **Doctors/Clinicians** – Review AI-generated pre-consultation reports, approve or modify triage actions, and view history.
- **Clinic administrators** – Track patient flow and pre-consultation efficiency via the dashboard (severity statistics, consultation history).

4. System Functionalities

4.1. Description: The system operates as a multi-stage decision engine focused exclusively on the pre-consultation phase. User input (unstructured text) is processed by an Ilmu GLM 5.1-based reasoning module that extracts medical entities (symptom, location, duration, severity). If information is insufficient, the AI generates targeted follow-up questions (one per turn) to complete the clinical picture. The Decision Engine then classifies risk into Low, Medium, or High and structures the findings into a clinical summary. The Workflow Engine selects the appropriate action, which is presented to a doctor-in-the-loop for approval. Once approved, the system generates a PDF summary for the consultation and displays an approval/rejection banner back to the patient

4.2. Key Functionalities:

- **Adaptive Patient Interviewing:** Translates unstructured patient language (free text) into structured medical entities (symptom, location, duration, severity) using Ilmu GLM 5.1. If information is insufficient, the AI generates targeted follow-up questions (one per turn) until the clinical picture is complete.

- **Risk Classification & Triage:** Processes extracted medical entities through a Decision Engine that classifies patient risk into Low, Medium, or High categories. Enables clinic staff to prioritize urgent cases while ensuring no critical symptoms are missed.

- **Doctor Review & Editing**: Presents the AI-generated clinical summary and risk classification to the attending doctor for review. Doctor can approve, reject, or modify the output before finalizing. An approval/rejection banner is displayed back to the patient for transparency.

- **PDF Summary Generation:** Converts the approved clinical insights into a structured, clinician-ready PDF document. The summary includes

4.3. AI Model & Prompt Design

4.3.1. Model Selection:

- **Model used:** Z AI GLM (ilmu-glm-5.1 via api.ilmu.ai).
- **Justification:** OpenAI-compatible endpoint, cost-effective for triage, multilingual support (English, BM, Chinese) and reliable JSON adherence.

4.3.2. Prompting Strategy: Our system uses multi-step agentic prompting with chain-of-thought reasoning, enabling adaptive follow-up questioning and step-by-step risk classification — essential for patient safety and complete data capture in a medical pre-consultation setting where Ilmu GLM 5.1 is not fine-tuned.

Multi-step agentic prompting with Chain-of-Thought reasoning allows the AI to ask one question at a time, collect all 20 required pieces of information, detect emergency situations immediately, and produce a well-organized report for doctors. This approach also reduces fake or incorrect outputs (hallucination) — which is especially important because our Ilmu GLM 5.1 model is not specifically trained on medical data.

4.3.3. Context & Input Handling: The system accepts unstructured text inputs up to ~4,000 characters and file attachments up to ~5MB — larger inputs are rejected with an error message, and no truncation or chunking is performed.

4.3.4. Fallback & Failure Behavior:

- **API & Network Failures (Graceful Error State):** When the AI model returns an error (network timeout, server error, or empty response), the system displays a user-friendly error message inside the chat bubble: *"Error: Unable to reach AI service. Please check your connection or try again."*
- **Off-Topic or Unusable Responses:** The system prompt explicitly instructs the AI to ignore non-medical inputs (greetings, jokes, off-topic questions) and redirect back to the next medical question. If the AI still returns an off-topic response, the system displays it as plain text.

- **Hallucinated Responses (Most Critical):** Forces AI to reason step-by-step before outputting clinical judgments by using chain-of thought prompting
- **Doctor-in-the-loop approval:** Every AI-generated report must be approved or rejected by a doctor before the patient sees it as final. The doctor can reject hallucinated or incorrect reports.

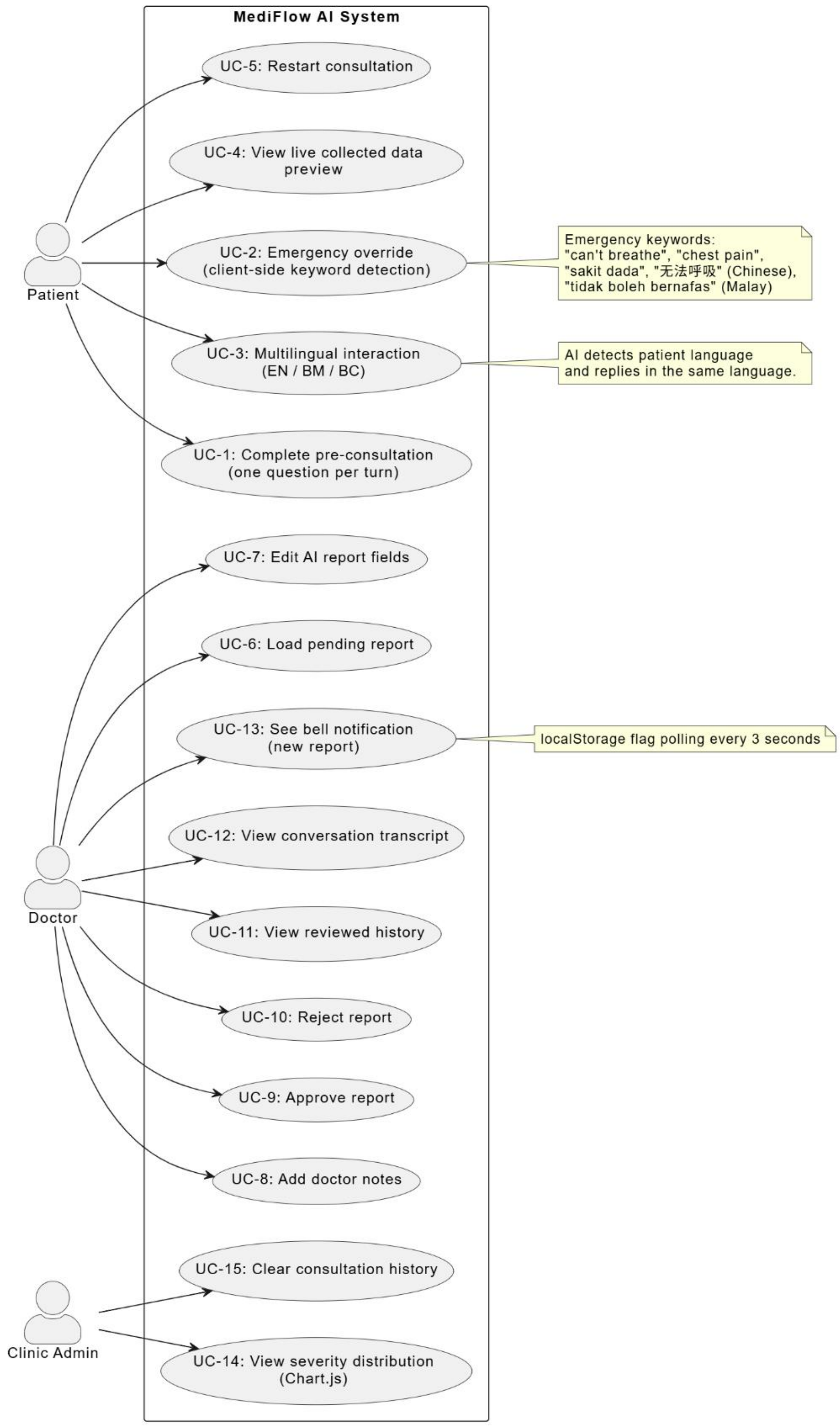
5. User Stories & Use Cases

- **User Stories:** As a patient, I want to describe my symptoms in my own words so that the AI can ask me relevant follow-up questions and generate a clear report summary that I can show to my doctor.
- **Use Cases (Main Interactions):**

Prompt to UI Flow: Patient enters symptoms in plain language, then MediFlow AI parses the intent, identifies missing medical entities, and asks one relevant follow-up question.

Flow Generation: System collects patient answers, then the GLM-based reasoning module extracts medical entities, classifies risk into Low/Medium/High, builds a structured JSON report, and saves it to browser localStorage for doctor approval.

- **Use Case Diagram:**



6. Features Included (Scope Definition)

- **Unstructured Input Processing**: Ability to parse casual/slang descriptions of pain or discomfort (supports English, Bahasa Malaysia, Chinese).
- **Intelligent Interviewing (Pre-consultation)**: AI-driven interactive Q&A to gather missing clinical data (one question per turn, validation logic, emergency override).
- **Report Generation**: Auto-generation of structured clinical reports for **pre-consultation** (not intra-consultation). Reports include chief complaint, location, duration, severity score, associated symptoms, medication history, possible causes, treatment plan, and AI confidence.

- **Decision Support Engine**: AI-generated suggestions for treatment plans based on identified risk levels, but all final actions require doctor approval.

- **Doctor Review Dashboard** – Full CRUD-like editing, approval/rejection, and history tracking using localStorage.

7. Features Not Included (Scope Control)

- **Inpatient/Post-Surgery Monitoring**: Real-time monitoring of hospitalised patients or nursing notifications.
- **Physical Vitals Integration**: Direct data fetching from hardware like blood pressure monitors or thermometers.

- **Rehabilitation Execution**: Physical tracking of exercises, diet, or medication intake (though a simple recovery task list is provided, it is not medically supervised).

- **Automatic Final Diagnosis**: The system does **not** diagnose; it only provides *suggestions* for human approval.

- **Mobile Push Notifications** – All notifications are browser-based (polling + badge).

- **PDF Export** – The current prototype exports data as JSON only (stored in localStorage). PDF generation is a future enhancement.
- **Dedicated Nurse Interface** – Nurses are not given a separate view; critical alerts are handled by the doctor dashboard.

8. Assumptions & Constraints

- **LLM Cost Constraint:** Estimation of token cost per average user session:
  A typical pre-consultation session collects 20 data points over approximately 10-15 conversation turns (user messages + AI responses). Each turn consumes:

Component	Estimated Tokens
System prompt (consultation mode)	~1,500 tokens
Conversation history (per turn)	~200-300 tokens
	~50-100 tokens
AI response	~100-200 tokens
Total per session	**~2,500 - 3,500 tokens**

- **Technical Constraints:**
  - The system uses browser local storage only. Data is not shared across devices or browsers. Reports and decisions are lost if。
  - browser cache is cleared. The prototype is built with vanilla HTML, CSS (Tailwind), and JavaScript. No support for React, Vue, Angular, or mobile frameworks (iOS/Android).
  - The AI chat function requires an active internet connection to reach the GLM API. Without internet, patients cannot start a consultation (though static pages still load).

- **Performance Constraints:**
    - The AI is configured with max_tokens: 2000 per response. Very long reports or complex answers may be truncated.
    - Maximum of 30 turns (MAX_TURNS: 30) to prevent infinite loops or excessive API costs. After 30 turns, the session may behave unexpectedly.
    - File attachments and chat rendering may be slower on low-end mobile devices due to base64 encoding and DOM manipulation.

- **User Input:**
    - AI cannot initiate a conversation. A patient must type an initial symptom description or question before any AI response is generated.
    - The system requires doctor-in-the-loop approval for all final reports. AI-generated summaries are not trusted as final clinical decisions and must be reviewed by a licensed doctor before patient use.
    - The system only accepts typed text input. Voice-to-text is not supported in the current prototype.

9. Risks & Questions Throughout Development

- **Hallucinated symptoms** – What if the AI invents symptoms the patient never mentioned (e.g., adding "fever" when patient only said "headache")?
- **Doctor availability** – What if no doctor is available to approve reports (after hours, weekends, or staff shortage)?
- **Incorrect medication advice** – If a patient asks about drug interactions, how do we prevent the AI from giving dangerous or incomplete answers?
- **Local storage clearing** – If a patient clears browser cache or uses incognito mode, their report is lost forever. How do we notify them?
- **Progress bar accuracy** – The progress bar assumes exactly 20 steps, but some patients may need fewer or more questions. Does this confuse users?